

LAB 1

PGM 1

//Write a program in C to find the second smallest element in an array.

```
#include<stdio.h>
```

```
int main(){
```

```
int n;
```

```
printf("enter the number of elements :");
```

```
scanf("%d",&n);
```

```
if(n<2){
```

```
    printf("insuffiecient elements");
```

```
}
```

```
int arr[n];
```

```
for(int i=0; i<n; i++){
```

```
    printf("Enter value:");
```

```
    scanf("%d",&arr[i]);
```

```
}
```

```
int smallest ,secsmallest;
```

```
//intializing smallest and second smallest val
```

```
if(arr[0]< arr[1]){
```

```
    smallest=arr[0];
```

```
    secsmallest=arr[1];
```

```
}
```

```
else{
```

```
    secsmallest=arr[0];
```

```
    smallest=arr[1];
```

```
}
```

```
for(int i=2;i<n;i++){
```

```
    if(arr[i]<smallest){
```

```
        secsmallest=smallest;
```

```

        smallest=arr[i];

    }
    else if(arr[i]<secsmallest&&arr[i]>smallest){
        secsmallest=arr[i];
    }
}
if(smallest==secsmallest){
    printf("no second smallest value");

}
else{
    printf("second smallest element %d",secsmallest);
}
}

```

PGM 2

//Write a program in C to find the sum of the left diagonals of a matrix.

```
#include<stdio.h>
```

```
int main(){
```

```
int rows,cols,i,j;
```

```
int sum=0;//initialising sum
```

```
//taking input for rows and columns of matrix
```

```
printf("enter number of rows:");
```

```
scanf("%d",&rows);
```

```
printf("enter number of cols:");
```

```
scanf("%d",&cols);
```

```
//checking if square matrix
```

```
if(rows!=cols){
```

```

        printf("Invalid matrix");
    }
    int matrix[rows][cols]; //creating 2d array for matrix
    //inputting the values for the matrix
    for(i=0;i<rows;i++){
        for(j=0;j<cols;j++){
            printf("enter element:");
            scanf("%d",&matrix[i][j]);
        }
    }
    //printing the matrix
    for(i=0;i<rows;i++){
        printf("\n");
        for(j=0;j<cols;j++){
            printf("%d ", matrix[i][j]);
        }
    }
    //finding sum of left diagonal
    for(i=0;i<rows;i++){

        sum+=matrix[i][rows-1-i];
    }
    printf("\n");
    printf("sum of the left diagonals:%d",sum);
}

```

PGM3

//Write a program in C to find the sum of rows and columns of a matrix.

```
#include<stdio.h>
```

```

int main(){
int rows,cols,i,j;
int rowsum=0;
int colsum=0;//initialising sum
//taking input for rows and columns of matrix
printf("enter number of rows:");
scanf("%d",&rows);
printf("enter number of cols:");
scanf("%d",&cols);
//checking if square matrix
if(rows!=cols){
    printf("Invalid matrix");
}
int matrix[rows][cols]; //creating 2d array for matrix
//inputting the values for the matrix
for(i=0;i<rows;i++){
    for(j=0;j<cols;j++){
        printf("enter element:");
        scanf("%d",&matrix[i][j]);
    }
}
//printing the matrix
for(i=0;i<rows;i++){
    printf("\n");
    for(j=0;j<cols;j++){
        printf("%d ", matrix[i][j]);
    }
}
//finding the sum of rows and columns
printf("\nSUM OF ROWS:");
for(i=0;i<rows;i++){

```

```

printf("\nsum of row %d :",i+1);
for(j=0;j<cols;j++){
    rowsum+=matrix[i][j];

}
printf("%d",rowsum);
rowsum=0; //set rowsum to 0 after each row
}
printf("\nSUM OF COLUMNS:");
for(i=0;i<cols;i++){
    printf("\nsum of column %d: ",i+1);
    for(j=0;j<rows;j++){
        colsum+=matrix[j][i];

    }
    printf("%d",colsum);
    colsum=0; //set column sum to 0 after each column
}
}

```

PGM4

//Write a program in C to count the total number of duplicate elements in an array.

```
#include<stdio.h>
```

```
int main(){
```

```
int n;
```

```
int count=0;//counter variable
```

```
//inputting the number of elements
```

```
printf("\nenter number of elements in the array:");
```

```
scanf("%d",&n);
```

```
int arr[n]; //creating array of size n
```

```
//taking input for elements of array
```

```
for(int i=0;i<n;i++){
```

```
    printf("\nEnter element:");
```

```
    scanf("%d",&arr[i]);
```

```
}
```

```
printf("array:");
```

```
for(int i=0;i<n;i++){
```

```
    printf(" %d",arr[i]);
```

```
}
```

```
printf("\n");
```

```
//comparing each element with all the other elements and incrementing count
```

```
for(int i=0;i<n;i++){
```

```
    for(int j=i+1;j<n;j++){
```

```
        if(arr[i]==arr[j]){
```

```
            count++;
```

```
        }
```

```
    }
```

```
}
```

```
printf("\nNumber of duplicates: %d",count);
```

```
}
```

